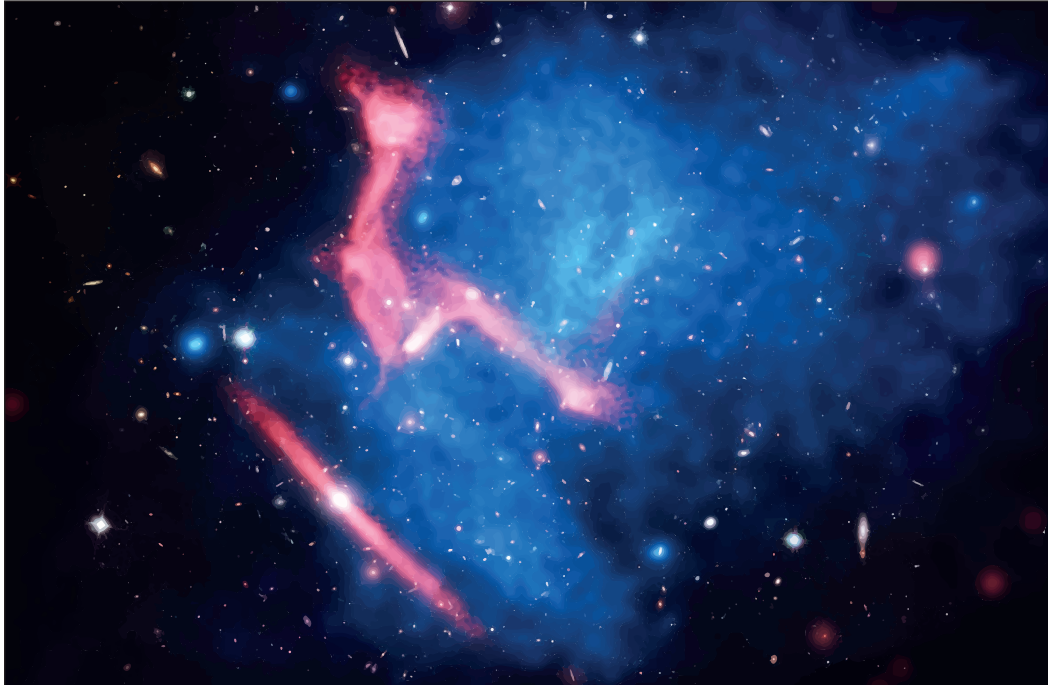


6. Three images have been combined below to show colliding galaxy clusters 5 billion light years from Earth. The background is a Hubble Space Telescope image taken using visible light; the blue area is an X-ray image from the Chandra Space Observatory, and the red area is an image from a radio telescope.



- (a) (i) Compare the use of Earth-based and space-based telescopes to produce images of objects in space. [2]

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- (ii) Explain how the visible light received by the Hubble Space Telescope provides evidence for the origin of the Universe. [3]

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- (b) (i) State how long it takes radio waves to travel from the colliding galaxy clusters to reach the Earth. [1]

time =

- (ii) X-rays have a higher frequency than radio waves. Tick (✓) the box next to the correct statement below. [1]

X-rays take less time to travel the same distance as radio waves.

☐

X-rays take the same time to travel the same distance as radio waves.

☐

X-rays take more time to travel the same distance as radio waves.

☐

- (iii) The X-rays detected by the Chandra Observatory have a frequency range of 3×10^{16} Hz to 3×10^{19} Hz. They travel through space at a speed of 3×10^8 m/s.

Use the equation

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the maximum wavelength of these X-rays. [3]

maximum wavelength = m

END OF PAPER

10





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1	H	Hydrogen	1
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7	Li	Lithium	3
9	Be	Beryllium	4
23	Na	Sodium	11
24	Mg	Magnesium	12
39	K	Potassium	19
40	Ca	Calcium	20
86	Rb	Rubidium	37
87	Fr	Francium	87

45	Sc	Scandium	21
48	Ti	Titanium	22
51	V	Vanadium	23
52	Cr	Chromium	24
55	Mn	Manganese	25
56	Fe	Iron	26
59	Co	Cobalt	27
59	Ni	Nickel	28
63.5	Cu	Copper	29
65	Zn	Zinc	30
89	Y	Yttrium	39
88	Sr	Strontium	38
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
99	Tc	Technetium	43
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
108	Ag	Silver	47
112	Cd	Cadmium	48
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
127	I	Iodine	53
131	Xe	Xenon	54
137	Ba	Barium	56
139	La	Lanthanum	57
179	Hf	Hafnium	72
181	Ta	Tantalum	73
184	W	Tungsten	74
186	Re	Rhenium	75
190	Os	Osmium	76
192	Ir	Iridium	77
195	Pt	Platinum	78
197	Au	Gold	79
201	Hg	Mercury	80
204	Tl	Thallium	81
207	Pb	Lead	82
209	Bi	Bismuth	83
210	Po	Polonium	84
210	At	Astatine	85
222	Rn	Radon	86

Key

relative atomic mass

A_r	Symbol	Name	Z
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atomic number